

The visit of Georg Schnabel (AIEA) at CEA Cadarache to incorporate TNC evaluation in gmapy was the opportunity to inspect the TNC and GMA data bases. For this work, we used the TNC values reported in the Axton report and the GMA values collected by Georg Schnabel in the “data2025.json” database. Some issues were identified and TNC database was corrected following suggestions of Roberto Capote (Table 1).

In addition it was decided to remove all total cross sections not measured with a metallic sample. Crystal lattice effect of oxide are taken into account via SCA parameter. Remove all datasets ABS(iso)+SCA(iso) and all quantities with SCA(iso).

Table 1: Summary of the changes in the TNC data base originally established by Axton.

Id Axton-TNC database	Id GMA database	Reference	Comments
70		Saplakoglu (1961)	• Change to 694.2(0.7%)
93		Lemmel (1982)	• Data connected to crystal extinction
95		Lemmel (1982)	• Deleted from TNC because large uncertainties
152 to 164		Cabel (1968)	• Deleted because T=116°C
54 to 61		Divadeenam (1984)	• Deleted because Westcoff factor (mac data not used)
76		Safford (1961)	• Pu239 data deleted as this is extrapolation
65		Nikitin (1955)	• Change ABS(35)+SCA(35) in ABS(35)+SCR(35) • Change to 710(2.8%)
73		Nikitin (1955)	• Change ABS(39)+SCA(39) in ABS(39)+SCR(39)
78		Nikitin (1955)	• Change to 580 barn
85		Simpson (1961)	• Change ABS(41)+SCA(41) in ABS(41)+SCR(41)
86		Craig (1964)	• Change ABS(41)+SCA(41) in ABS(41)+SCR(41) • Change to 1383(3.0%)
79		Patenden (1956)	• Change to 590 barn
	8026	Adamchuk (1988)	• Add Adamchuk data in TNC (1 data set) • Remove in GMA data base
	1918	Arif (1987)	• Add Arif data in TNC (1 data set) • Remove in GMA data base
	8099	Lounsbury (1970)	• Values verified by Roberto Capote
	8098	Lounsbury (1970)	• Added in TNC, Removed from GMA
	8097	Lounsbury (1974)	• Lounsbury (1974) originally from 1970.
		Walner (2014)	• Add Walner $^{235}\text{U}(n_{\text{th}}, \gamma)$ values (2 data sets) • See PRL 112, 192501 (2014)
			• Remove all total cross section ABS(iso)+SCA(iso) • Remove SCA(iso) values • Remove all quantities with SCA(iso)

Before the final gmapy analysis:

- Update half-life of Au197, Co59, Mn55, B10, S, Bnat
- Verify consistency between Boron and Gold thermal values between TNC and GMA database
- Update uncertainties of $v_{\text{tot}}(\text{Cf252})$ according to discussions organized by Denise

It is also recommended to add in the final standard files:

- Thermal delayed neutron multiplicities (See CRP of Vivian) and the deduced thermal v_p
- Thermal capture cross section of ^1H used by Gerry (± 200 pcm of uncertainty on PWR)

Other Data sets and corrections are proposed below (Robert Capote, 17/09/2025)

Add scattering data:

- SCR(33) OLEKSA 1958 - Phys Rev.109,1645(1958)- 12.5 (4%)
- SCR(33) Green 1974- 12.30 (5.6%)
- SCR(33) VERTEBNIY 1974- 13.2 (2.2%)
- SCR(33) BLOCK 1960- 11(18%)
- SCR(35) BLOCK 1960- 13(15%)
- SCR(35) CEULEMANS 1970- 14.3 (3.5%)
- SCR(35) FOOTE 1958- 15.0 (6.0%)
- SCR(39) Safford 1961 11(30%)

Please also add/change the following data:

- ABS(33) + SCR(33) Pshenichnyj 1978- 587.9 (0.5%)
- ABS(33) + SCR(33) Harvey 1979- 590 (0.8%)
- ABS(33) + SCR(33) Moore 1979- 587 (1.0%)
- **ABS(33) + SCR(33) Block 1960- 587 (0.5%) Id=84**
- **ABS(33) + SCR(33) Safford 1960- 587 (0.85%) check Id=83**
- **ABS(33) + SCR(33) Pattenden 1956- 590 (2.5%) Id=79**
- ABS(35) + SCR(35) Spencer 1987 690 (0.8%)
- ABS(35) + SCR(35) Leonard 1954- 702 (1.0%)
- **ABS(35) + SCR(35) Safford 1959- 698.3 (0.3%) Id=68**
- **ABS(35) + SCR(35) Simpson 1960- 690 (1.4%) Id=66**
- ABS(35) + SCR(35) Antonov 1986- 695 (2.4%)
- ABS(35) + SCR(35) Guerasimov 1962 687 (1.2%)
- **ABS(35) + SCR(35) Saplakoglu 1961- 694.2 (0.7%) Id=70**
- **ABS(35) + SCR(35) PALEVSKY 1954- 700 (0.7%) Id=64**
- ABS(39) + SCR(39) Anderson 1945- 1045 (2.4%)
- ABS(39) + SCR(39) Havens 1954- 1067 (1.9%)
- ABS(39) + SCR(39) Palevsky 1955- 1034 (1%) update
- ABS(39) + SCR(39) Leonard 1956- 1055 (1.3%)
- **ABS(39) + SCR(39) Bollinger 1958- 1030 (1.0%) update Id=74**
- ABS(39) + SCR(39) Stoughton & Halperin 1959- 1030(3.9%)
- ABS(39) + SCR(39) Spencer 1987- 1025 (0.6%)